Autonomous Solar-Powered Spot Heater

Critical Off-Grid Thermal Protection

PIC-G21 · ElectroCap – 2026

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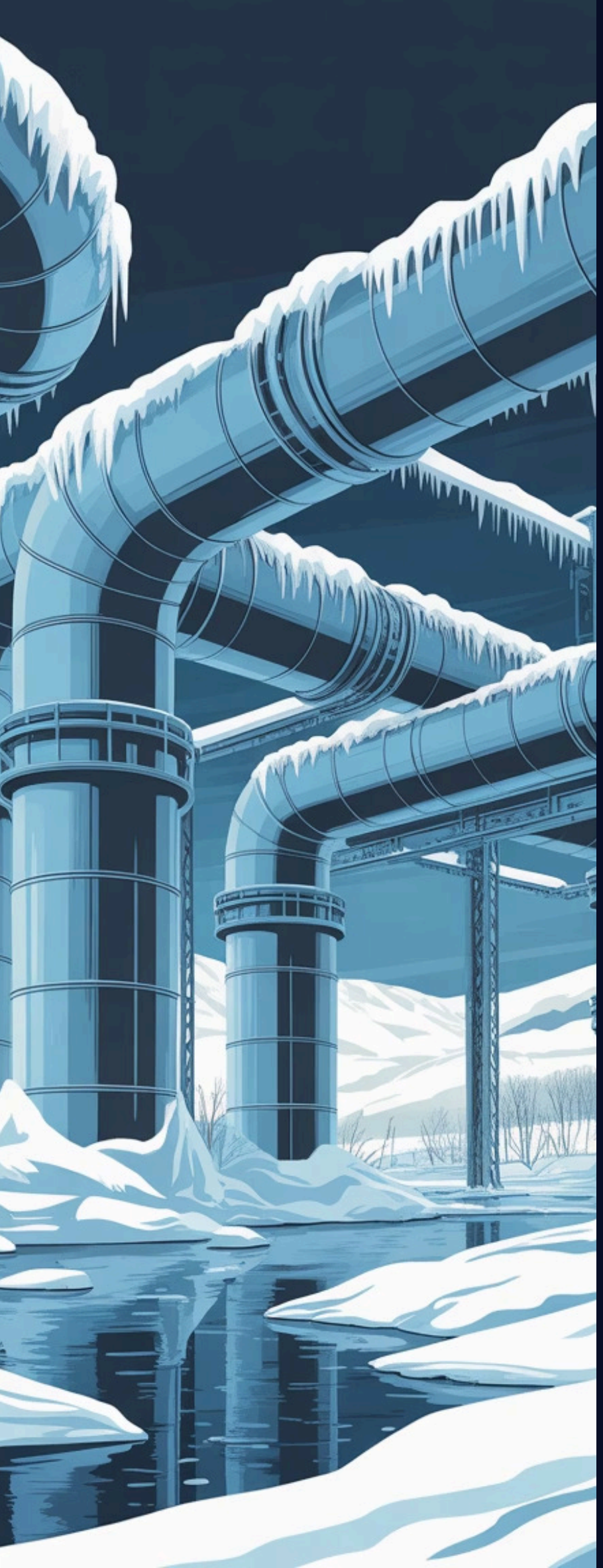

CHAPTER 1 THE PROBLEM

The Problem

Critical infrastructures in remote cold environments fail due to temperature.

Frozen Pipes

Ruptured tanks and burst pipes

Battery Collapse

Capacity drops below 0°C

Comms Breakdown

Communication systems fail

Costly Repairs

Emergency maintenance expenses

> **There is no simple, autonomous, low-power spot heating solution for off-grid assets.**

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Market Opportunity

Growing demand for resilient off-grid infrastructure driven by **decentralization + climate volatility + asset resilience**.

Renewable Energy Sites

Remote Telecom Stations

Agriculture & Apiaries

Environmental Monitoring

Civil Protection Systems

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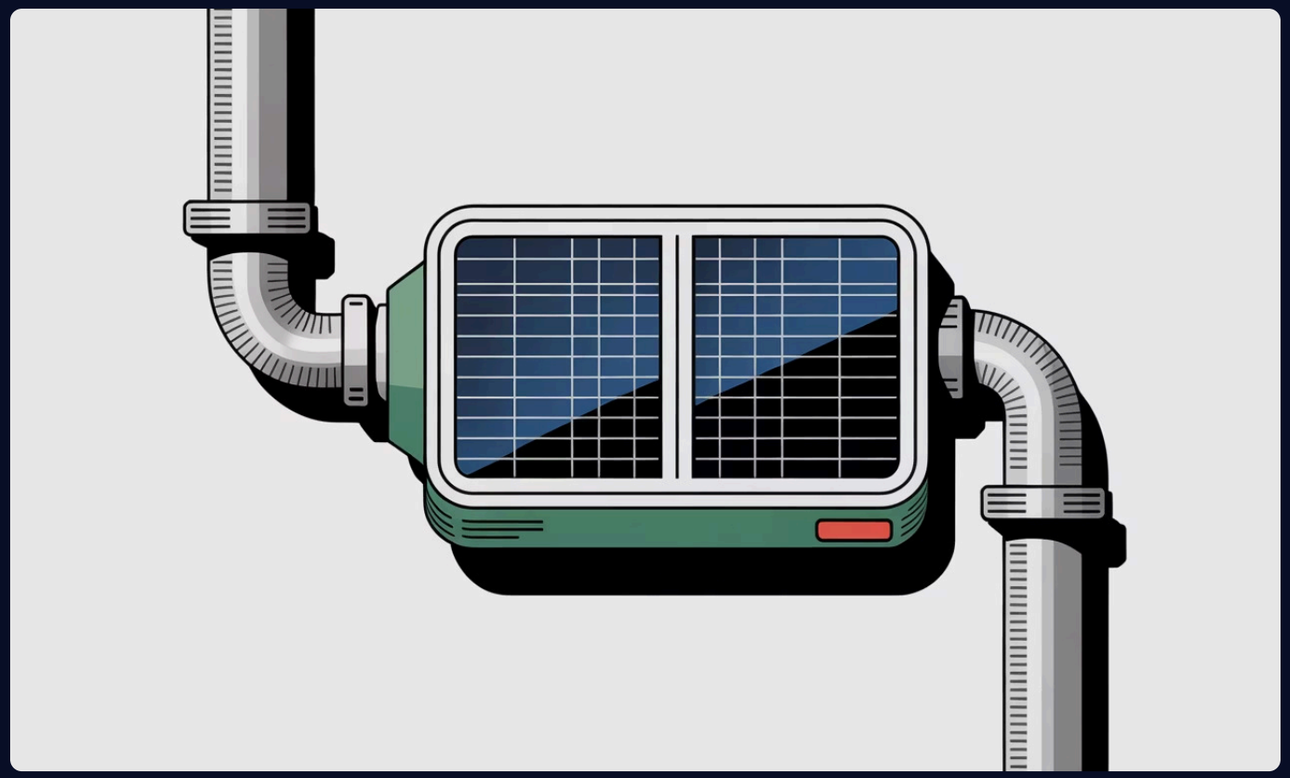
Our Solution

A fully autonomous solar-powered spot heating system. **We heat only what matters.**



01

Off-Grid PV Supply

02

Intelligent MPPT Energy Management

03

Battery Storage with Protection

04

Smart Thermal Control (30–50W)

05

Optional Remote Monitoring (LoRa/Wi-Fi)

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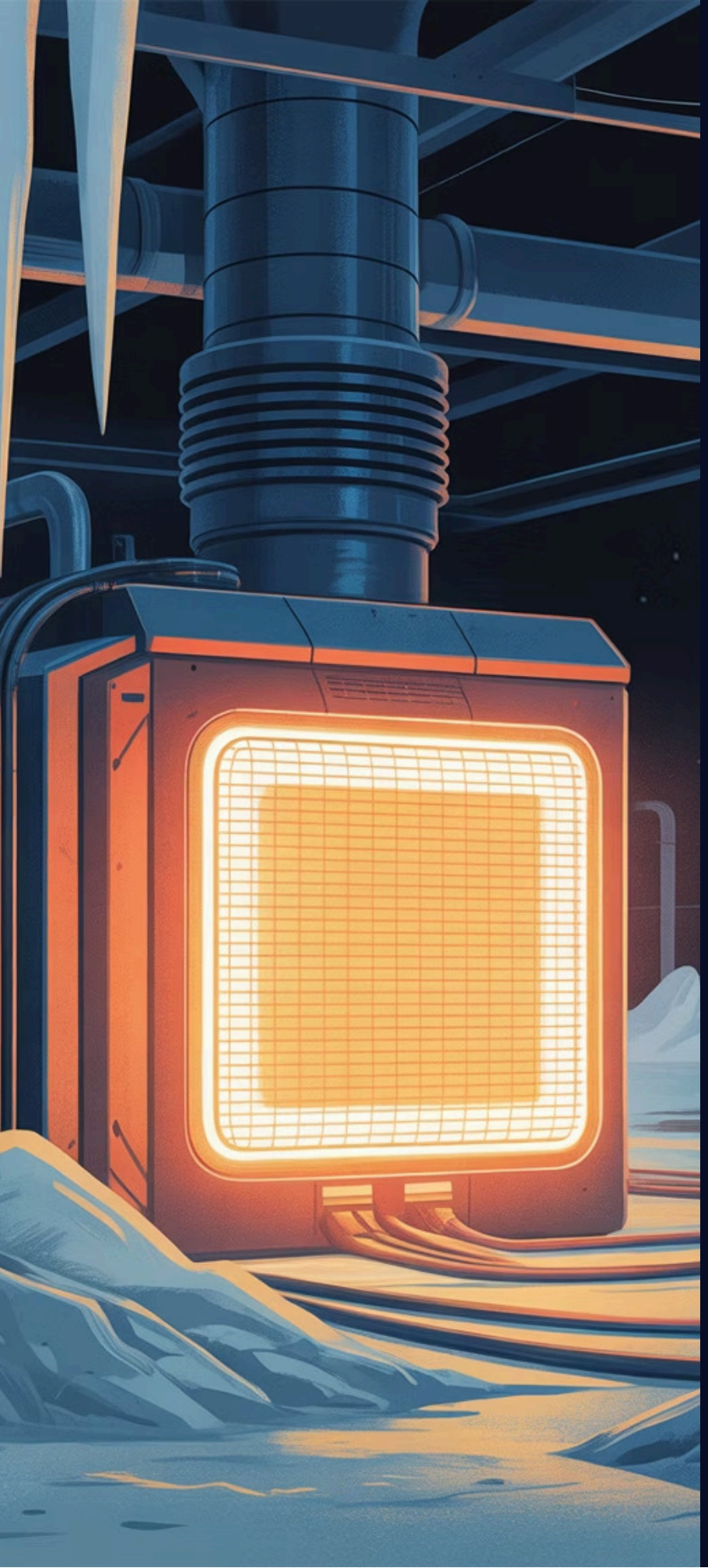

CHAPTER 4 VALUE PROPOSITION

Why We Are Different

Ultra Low Power

30–50W consumption

Localized Heating

Spot heating, not full enclosure

Multi-Day Autonomy

Solar-powered resilience

Modular & Scalable

Low install & maintenance cost

Resilience without grid dependency.

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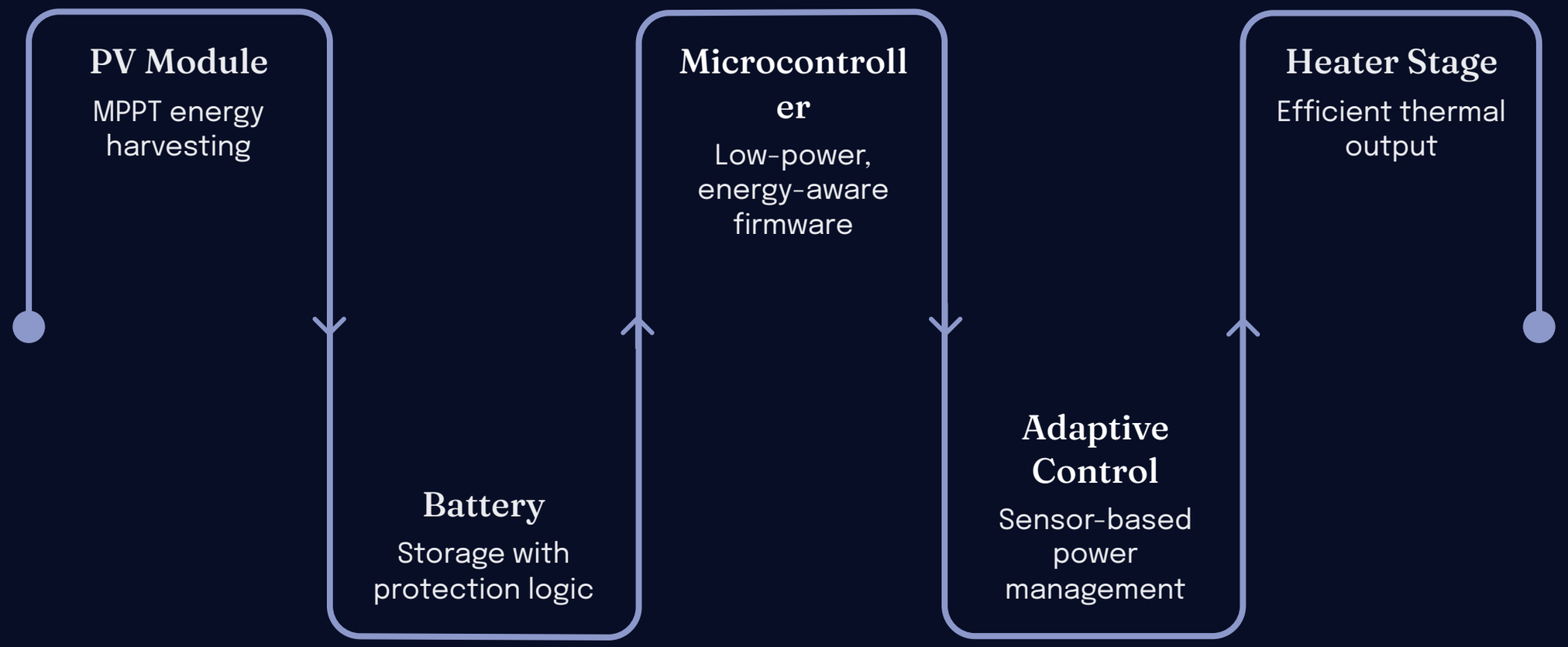
Technology Architecture

Integrated system with energy-aware firmware for maximum autonomy.



Each module communicates through an energy-aware firmware layer that dynamically balances harvesting, storage, and consumption.

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Competitive Landscape

Alternative	Autonomous	Intelligent
Manual (blankets, gas)	❌	❌
Grid-connected industrial	❌	Partial
DIY battery heaters	Partial	❌
Niche solar beehive systems	✅	❌
PIC-G21 (Ours)	✅	✅

Innovation gap: No modular, intelligent, low-power, asset-focused solution exists on the market today.

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Technical Challenges

PV & Battery Sizing

Worst-case winter scenarios

Low-Temp Battery Mgmt

Safe operation below 0°C

Heater Drive Electronics

Efficient power delivery

Thermal Control Algorithms

Stable, adaptive regulation

Ultra-Low-Power Firmware

Maximized energy autonomy

We address these with **integrated energy modelling** and **robust embedded control**.

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CHAPTER 8 VALIDATION

Validation Metrics

Data-driven validation, not qualitative claims.

Multi-Day Autonomy

Operation without sunlight

Temp Accuracy

$\pm 2^{\circ}C$ control precision

Energy Ratio

Harvesting vs. consumption balance

Fault Tolerance

Safe-state transitions

Comms Reliability

If remote monitoring enabled

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Execution Plan

Structured development roadmap with an **iterative build-test-improve cycle**.

1. **Requirements**
   Definition & scoping
2. **Hardware**
   Design & assembly
3. **Firmware**
   Embedded development
4. **Module Testing**
   Component-level validation
5. **System Validation**
   Integrated testing
6. **ElectroDay**
   Final prototype demo

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